

2024



Modernizing Applications in Minutes Using Amazon Q Developer with Novacomp

Learn how Novacomp accelerated legacy Java applications modernization, improved security, and reduced technical debt by an average of 60 percent.

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60%



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reduction on average in technical debt

50 minutes

10,000 lines of code upgraded in 50 minutes instead of 3 weeks

Overview

As a leading IT company in Latin America, [Novacomp](#) serves a portfolio of clients across many industries, many of whom are increasingly looking to build solutions in the cloud. To facilitate the adoption of cutting-edge technologies, in 2023, Novacomp launched a new division focused on assisting clients with the modernization and optimization of their operations using Amazon Web Services (AWS).

Historically, the modernization of applications has been a time-consuming task that steals time away from the development of new features and other important initiatives. Novacomp decided to incorporate generative artificial intelligence (AI) to alleviate the challenges of keeping its operations up to date. In particular, it turned to [Amazon Q](#), a generative AI-powered assistant designed for work that can be tailored to an individual business.

Using [Amazon Q Developer](#), the most capable generative AI-powered assistant for software development, Novacomp upgraded a project with more than 10,000 lines of Java code in 50 minutes rather than an estimated 3 weeks. The company also simplified everyday tasks for developers, reduced its technical debt by 60 percent on average, and helped customers to significantly improve their security posture and save associated costs.



Opportunity | Using Amazon Q Developer to Modernize Legacy Java Applications for Novacomp

Founded in Costa Rica in 1998, Novacomp operates in 11 countries, supporting hundreds of companies with outsourcing, quality assurance and software testing, software development and applications, and IT consulting. Increasingly, Novacomp's clients are looking to migrate to AWS, implement the latest technologies, build solutions in the cloud, and incorporate AI.

But Novacomp engineers were spending a disproportionate amount of time on maintenance of the company's base code, much of which was written in Java 8 and was becoming obsolete. Specifically, Novacomp sought to upgrade 10,000 lines of code from one project written in Java 8 to Java 17, a task that the company estimated would take a senior Java architect at least 3 weeks to complete.

In 2023, Novacomp attended [AWS re:Invent](#) and saw a demonstration of Amazon Q Developer. "The very minute I saw that demo, that was a game changer for me," says Gerardo Arroyo, Novacomp's chief technology officer for cloud. "I wondered what would happen if we migrated these 10,000 lines of code from Java 8 to Java 17 using Amazon Q Developer." As an assistant powered by generative AI, Amazon Q Developer has trained on 17 years of AWS information and knowledge, including documentation, blog posts, software development kits, APIs, and open-source data and code. Novacomp recognized how using Amazon Q Developer simplifies time-consuming tasks such as patching, updating programming languages, and upgrading dependencies—when one software component relies on another to work properly. Because many of Novacomp's clients operate in the financial sector, the company also wanted to use Amazon Q Developer to prioritize best practices in security.

Solution | Upgrading 10,000 Lines of Code in 50 Minutes Using Amazon Q Developer Agent for Code Transformation

To update its legacy code, Novacomp turned to [Amazon Q Developer Agent for code transformation](#), which accelerates application maintenance, upgrades, and migration in minutes. The company automatically upgraded 10,000 lines of Java in 50 minutes rather than the expected 3 weeks. "Using Amazon Q Developer Agent for code transformation, we took the code, recompiled everything, deployed it, and it worked perfectly the first time," says Arroyo. The upgrade to Java 17 also reduced vulnerabilities in legacy code, lowering Novacomp's risk by an average of 60 percent. "By upgrading legacy code to the latest version, we address our clients' compliance issues that include security and vulnerabilities and save them costs in the long run. That's a great side effect of using Amazon Q Developer Agent for code transformation."



In addition to the code upgrade, Novacomp connected Amazon Q Developer to its internal knowledge bases to streamline everyday tasks and complete complex development tasks faster. Developers ask questions using natural language to find accurate and up-to-date information about relevant AWS services, generate code, troubleshoot errors, and establish architectural best practices for Novacomp applications on AWS. “Using Amazon Q Developer, we can write a lot of applications in hours, with good precision and quality,” says Arroyo. “In the end, that saves money for ourselves and for our clients.”

Additionally, Novacomp uses Amazon Q Developer to transform how developers onboard and become productive faster. Novacomp is hiring developers at a pace of nearly one per day, with more than 1,000 developers in total. These new hires need to unravel thousands of lines of the company’s base code right away so that they can quickly understand what programs do and work on applications for clients. “Our new developers can ask Amazon Q Developer any question regarding that base code and get a clear, precise answer in natural language,” says Arroyo. “The developers can then create or fulfill new requirements with ease. And that accelerates the number of tickets that they can resolve in a given amount of time.”

Outcome | Reallocating Time and Cost Savings to Accelerate Development Pipelines

Using Amazon Q Developer Agent for code transformation, Novacomp has migrated about 80 percent of its own base code to the latest version of Java, and it’s helping some of the biggest public sector companies in Costa Rica to upgrade. The company is reallocating its time and cost savings to modernizing additional workloads for clients and building new solutions on AWS. In addition to new custom software development, the company is exploring new initiatives, such as the implementation of [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing foundation models from leading AI companies.

“Due to the benefits of Amazon Q Developer and the rapid modernization of code, our highly specialized Java senior developers can work on other projects for our clients,” says Arroyo. “That’s quite important, because we are accelerating the pipeline of projects.”

About Novacomp

Based in Costa Rica, Novacomp is a digital leader in the development, maintenance, and testing of software solutions, helping its clients to maximize their efficiency and profitability.

AWS Services Used

Amazon Q

Amazon Q is the most capable generative AI-powered assistant for accelerating software development and leveraging companies' internal data.

[Learn more »](#)



Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon through a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

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